

27/07/26



1.) $T(n) = T(n-1) + n \log n$. Step by Step derive Complexity.

A)

$$T(n) = T(n-1) + n \log n$$

$$= T(n-2) + \log(n-1) + n \log n$$

$$= T(n-3) + \log(n-2) + \log(n-1) + n \log n$$

$$T(n) = T(1) = \sum_{i=2}^n \log(i)$$

$$\sum \log i = \log(n!)$$

$$\log(n!) = \Theta(n \log n)$$

$$T(n) = \Theta(n \log n)$$

- Interpolation This algorithm is insufficient Extremely
- Large Volume of Large Security Threat Cost...

Unit-1

CSA-0606

Design and Analysis Algorithm

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2.) $T(n) = 5T(n/2) + n$. Master Theorem.

A) $a=5$ $b=2$, $f(n)=n$

$$n \log b^a = n \log_2 5$$

$$\log_2 5 = 2.32$$

$$n \log_2 5 = n \cdot 2.32$$

$$n < n^{2.32}$$

$$T(n) = \Theta(n \log_2 5)$$

$$\Theta = (n^{2.32})$$

Justification: Since $f(n)$ grows polynomially slower than falls

into Case 1 of Master Theorem.

Thus complexity Δ performance

Find Time Complexity $\Theta(n \log n)^{2.32}$

Performance Impact: A complexity of $\approx O(n)$ scales poorly

For very large distributed system.

$$3.) T(n) = 2T(n/2) + n^2 \log n \quad \text{Compare}$$

$$n \log_a \dots$$

$$A) a=2, b=2$$

$$n \log_2 2 = n$$

$$n^2 \log n \geq n$$

$$n \log b^a = n \log_2 2 = n$$

$$\text{Compare } f(n) = \text{with } n \log b^a$$

$$\text{Compare } f(n) = \text{with } n \log b^a$$

$$f(n) = n^2 \log n = \Omega(n^{1+\epsilon}) \quad \text{where } \epsilon=1$$

$$\text{Check } (c.F)(n,b) \leq c F(n) \quad \text{For } c < 1;$$

$$\left(\frac{1}{2}\right)^2 \log\left(\frac{n}{2}\right) = \frac{1}{2} n^2 (\log - \log^2) \leq 2 n^2 \log n$$

This Condition where true $\geq c \frac{1}{2}$

* Justification Since $f(n)$ grow polynomially faster and satisfied

Regularity.

* Falls into case Three Master Theorem.

$$2.) T(n) = 3T(n/3) + 1$$

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A) $T(n) = 3T(n/3) + 1$

$$T(1) = c$$

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$$T(n) = 9T(n/9) + 3 + 1 \quad 3^2 T(n/3^2) + 3^1$$

$$T(n) = 9 [3T(n/27) + 9 + 3 + 1]$$

$$T(n) = 27T(n/27) + 9 + 3 + 1 + 3^2 (n/3^2)$$

$$T(n) = 3^k T(n/3^k) + \sum_{i=0}^{k-1} 3^i$$

$$\sum_{i=0}^{k-1} 3^i = \frac{3^k - 1}{3 - 1} = 3^k - 1$$

$$\frac{n}{3^k} = 1 \Rightarrow n = 3^k \Rightarrow k$$

$$T(n) = nT(1) + \left(\frac{n-1}{2} \right)$$

$$T(n) = \frac{1}{2}n + \frac{1}{2}$$

$$T(n) = c + 1 = 1$$

$$T(n) = \Theta(n)$$

Asymptotic Complexity:

The domination term is approaches insufficient it is

Find Bound $T(n) = \log_3(n)$ Limit Time Complexity.